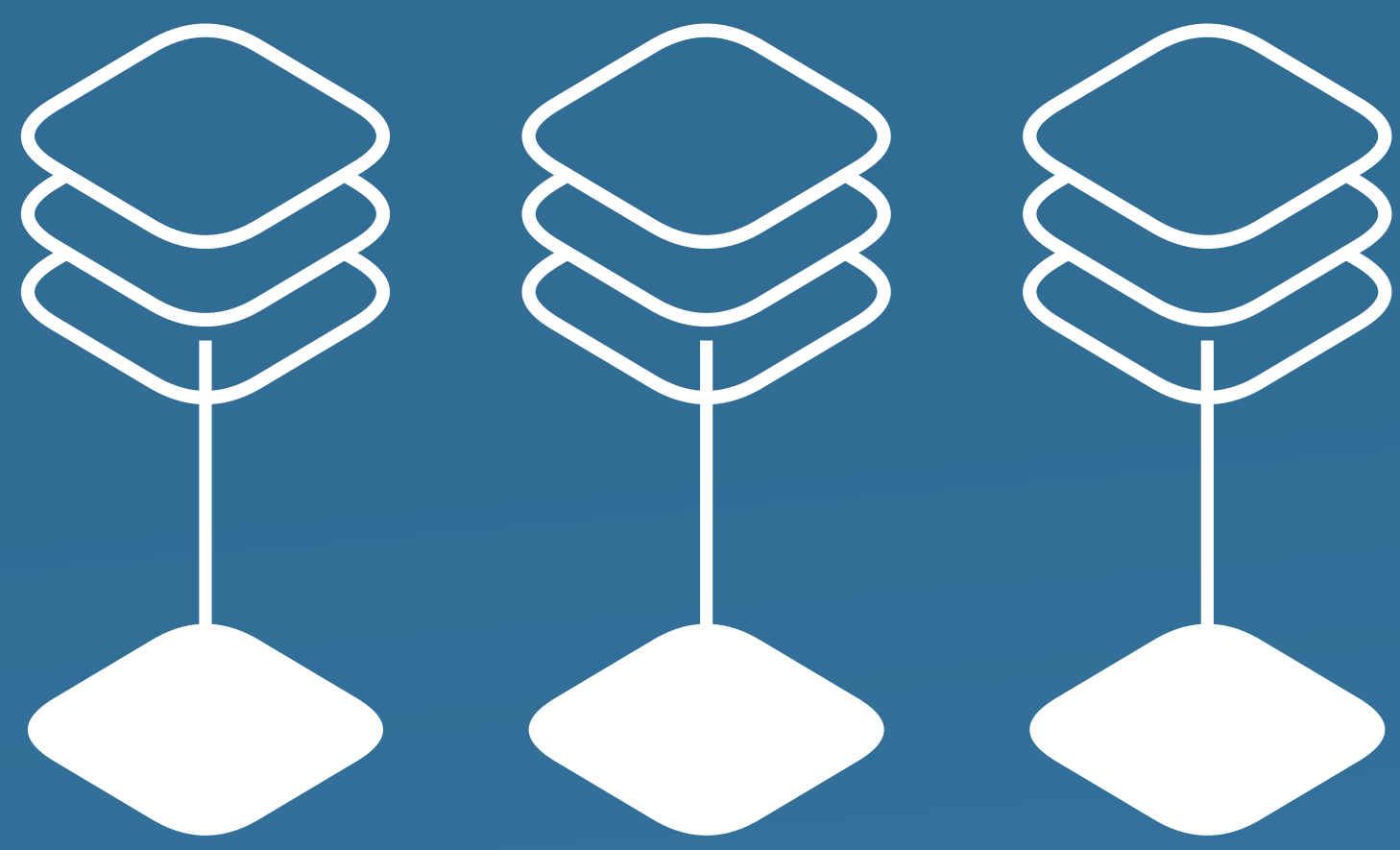
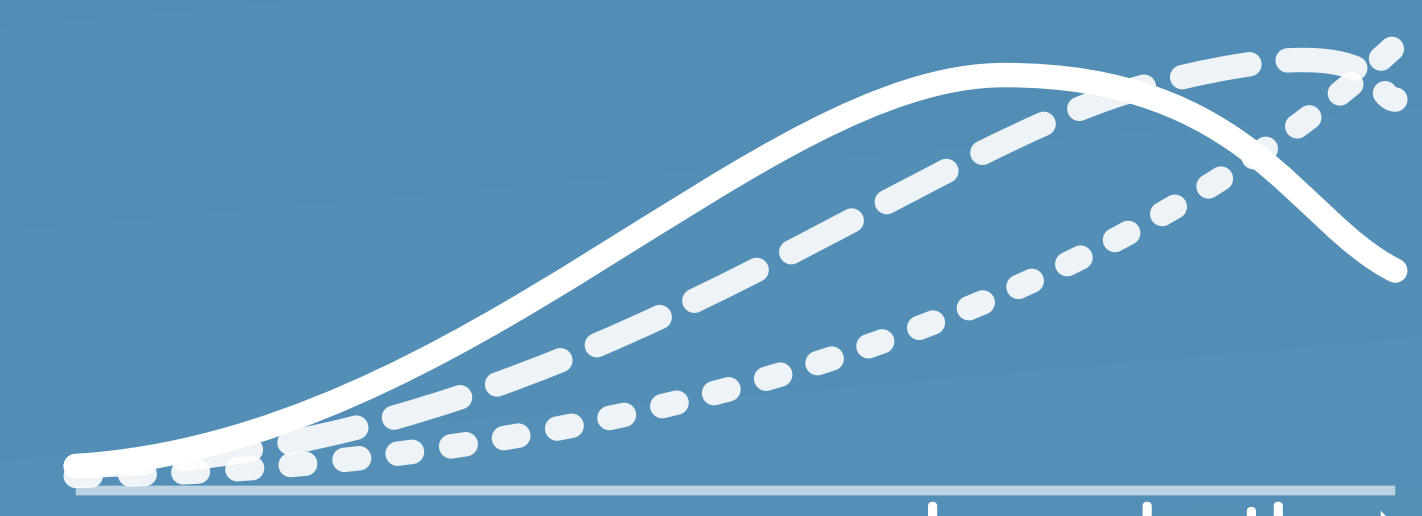




Which layer of a frozen vision foundation model should you probe?

None alone.

For dense prediction, try **AMDF**: it fuses **all** layers, for **every** patch.



On DINOv3-L, each NYUv2 task peaks at a different depth

**≈22×**

fewer params than concatenating layers (ViT-L, dense tasks)



Let's talk!

Open to research collaborations

vision foundation models · representation learning · probing & interpretability · spatial understanding



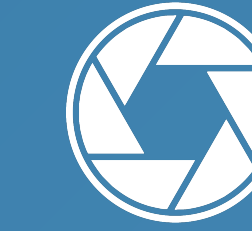
turhancankargin.me

# AMDF: Holistic Probing of Vision Foundation Models with Adaptive Multi-Layer Dense Fusion

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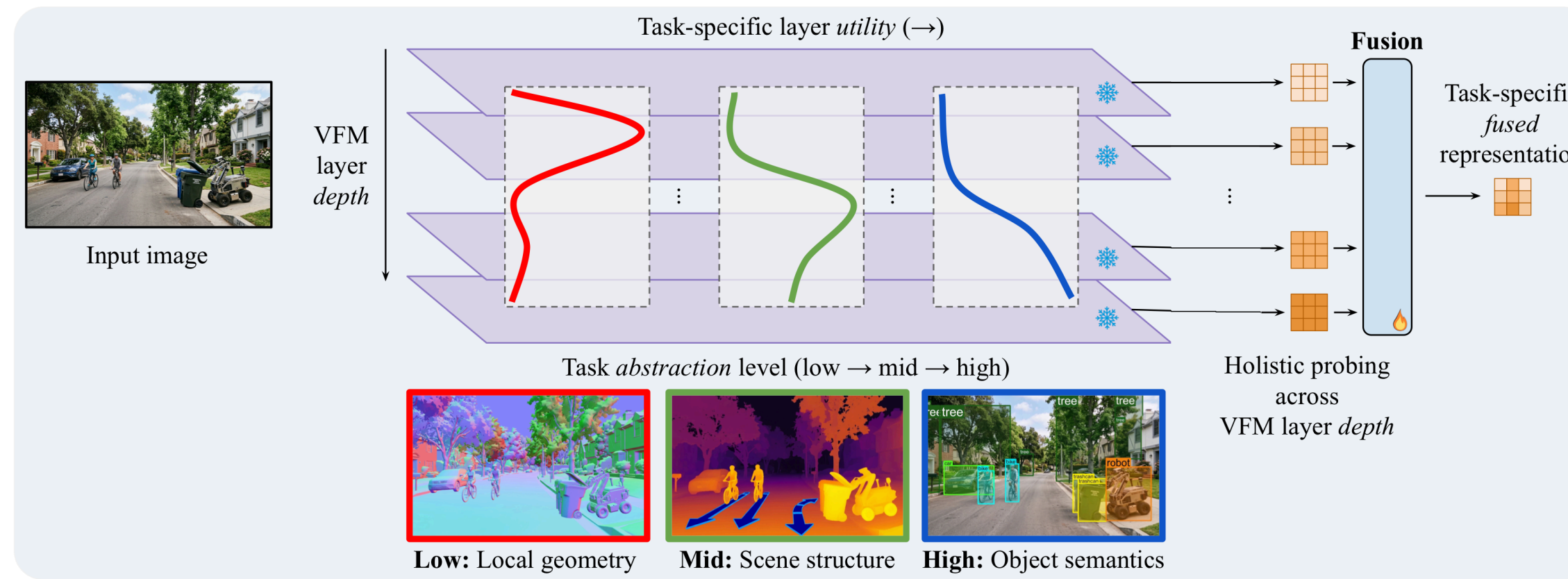
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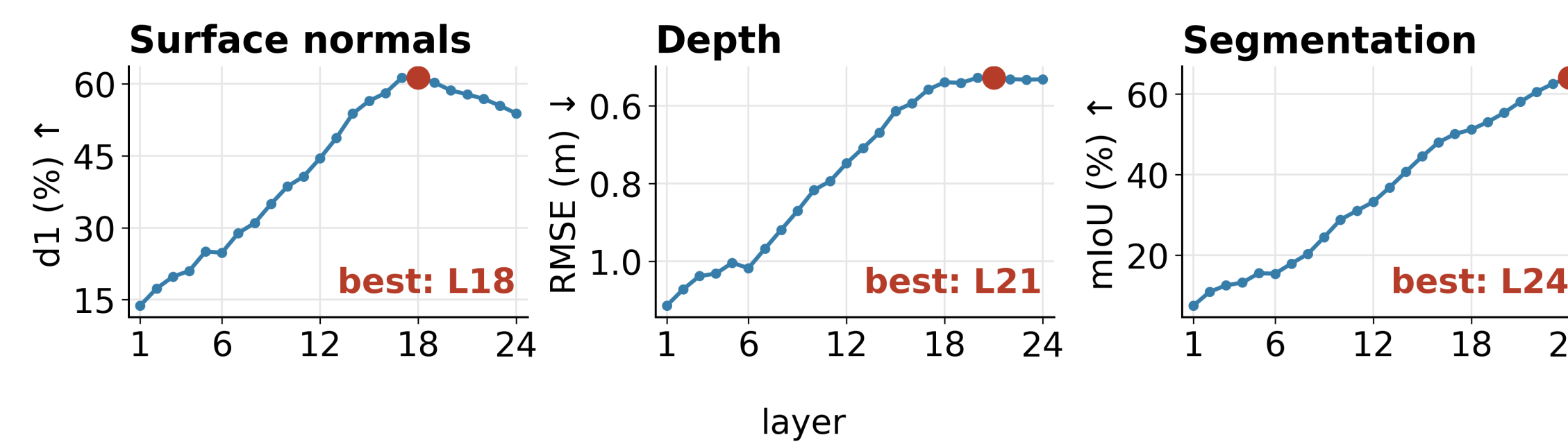
Copernicus Science Centre  
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## Tasks live at different depths

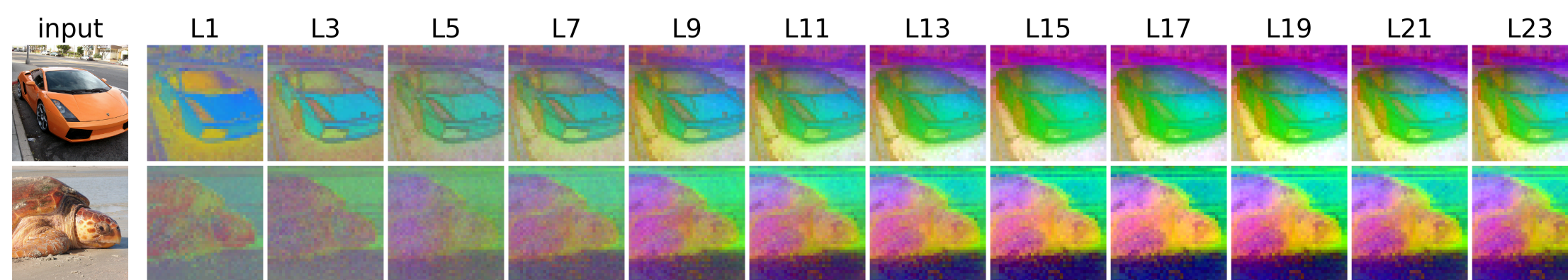


**Fig. 1:** Geometry peaks mid-depth, semantics at the end; AMDF fuses all layers per patch.

## #1: last layer ≠ always best



**Fig. 2:** Best single layer differs per task: ≈L18 (normals), ≈L21 (depth), L24 (segmentation); one seed per layer (DINOv3-L, NYUv2).



**Fig. 3:** MAE-L, ImageNet: **fine detail** early, **object regions** and parts (turtle shell vs head) deep.

## #2: all layers = costly

**✗** Concat all layers: **26.2M** params, the same **fixed weights** for every patch.

**✓** AMDF: **per-patch** layer choice, **≈1.2M** params.

## Take-home

- Useful features are **spread across depth**.
- AMDF: **2nd-best mean rank** with **≈22×** **fewer params** than Concat (ViT-L).
- AMDF beats every **fixed layer mix** by mean rank.

7 frozen VFMs

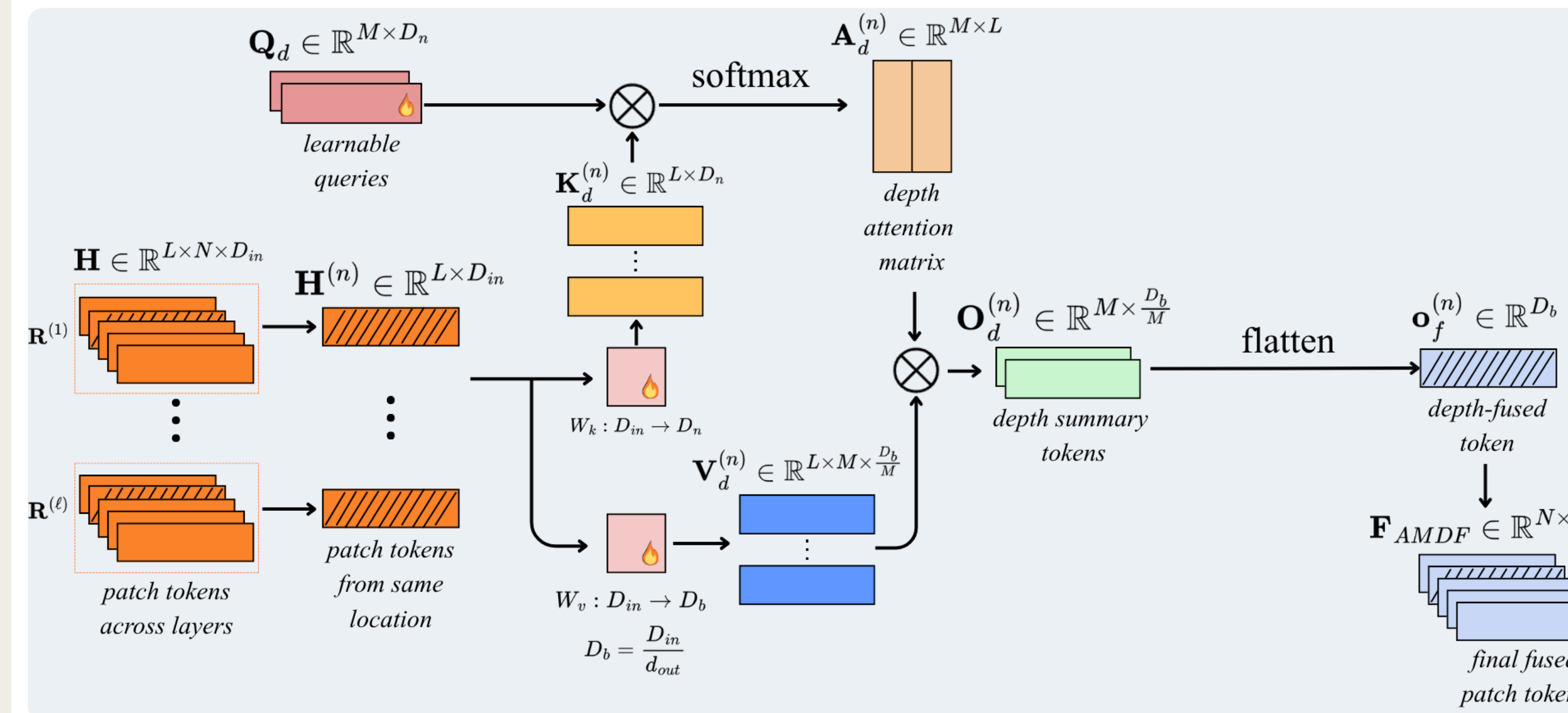
5 benchmarks

5 baselines

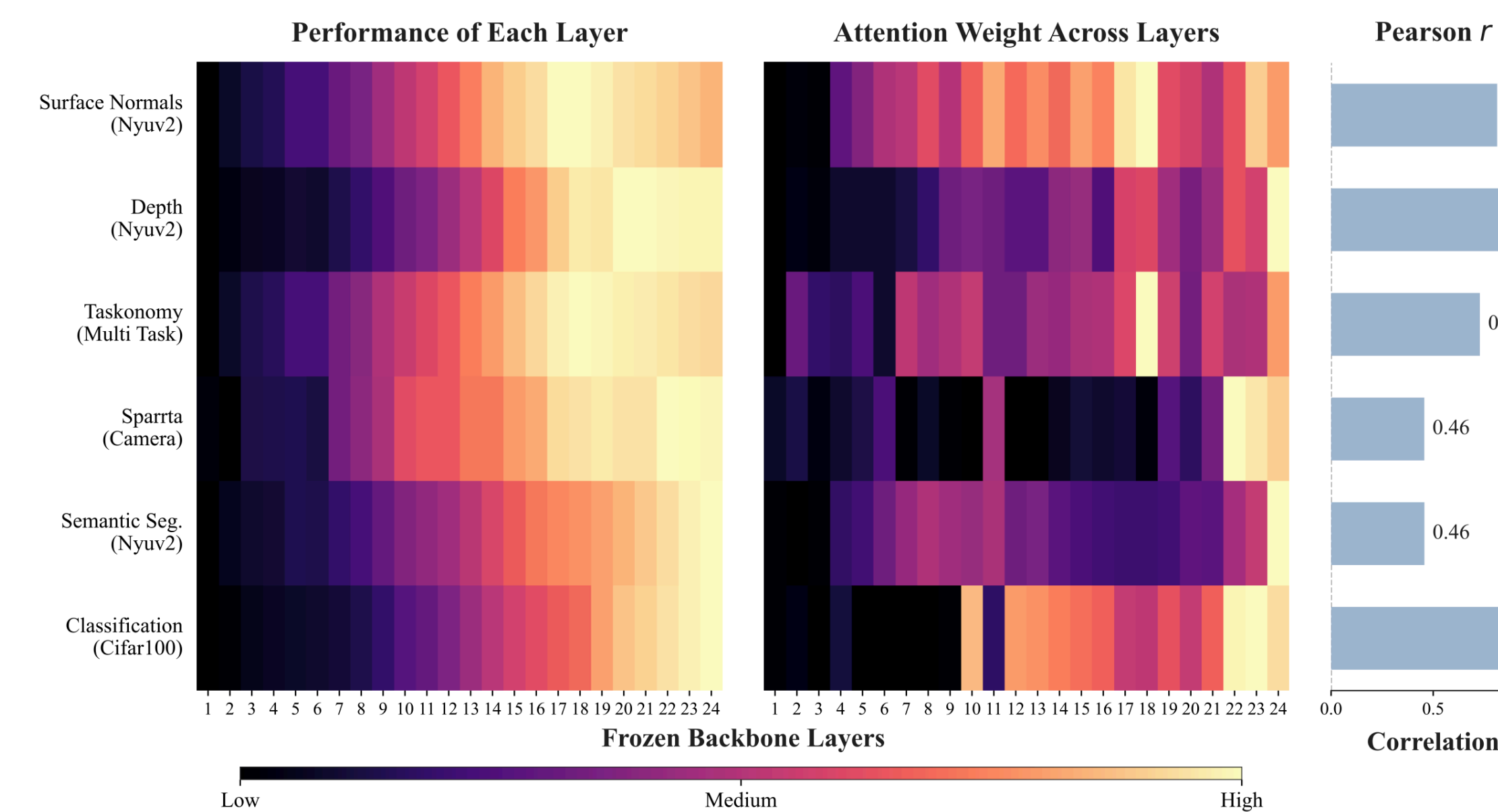
3 seeds

## Solution: AMDF — depth-aware fusion

- Per patch, learnable queries score all layers
- Softmax over depth → a task-specific layer mix
- Frozen backbone, light task heads

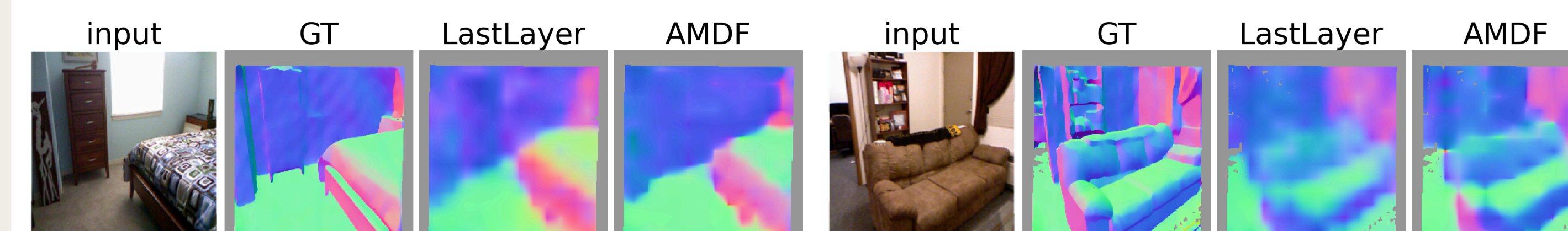


## Attention vs. layer utility (earlier variant)



**Fig. 4:** Earlier AMDF variant (DINOv3-L): attention correlates with per-layer probe utility,  $r = 0.46$ – $0.85$  across 6 tasks.

## What AMDF predicts



**Fig. 5:** NYUv2 normals, DINOv3-L (seed-8 retrain), selected examples (gain at the 83rd / 52nd percentile): AMDF recovers **sharper geometry** than the last-layer probe.

## Per-layer queries: better mean on all 5 tasks

| Normals       | Taskonomy     | Segm.         | Depth         | Pose          |
|---------------|---------------|---------------|---------------|---------------|
| +1.62 d1      | -0.73 L1      | +0.16 mIoU    | +0.51 d1      | -2.37° rot.   |
| 7/7 backbones | 7/7 backbones | 6/7 backbones | 5/7 backbones | 5/7 backbones |

Paired gains vs. shared queries, mean over 7 backbones; Taskonomy L1 ×100; cost: +2% params.

## Results

**≈22×** fewer params than Concat (ViT-L)

**2nd** mean rank (1st on depth)

**5×7** benchmarks × backbones

## Mean rank per benchmark

|                    | Normals       | Depth         | Segm.         | Taskonomy     | Pose          |
|--------------------|---------------|---------------|---------------|---------------|---------------|
| <b>AMDF (ours)</b> | 2.00          | <b>1.43 ★</b> | 1.86          | 2.00          | 1.93          |
| ConcatLayers       | <b>1.00 ★</b> | 1.71          | <b>1.57 ★</b> | <b>1.00 ★</b> | <b>1.21 ★</b> |
| WeightedLayers     | 4.14          | 4.00          | 4.14          | 3.29          | 3.89          |
| SpatialWeights     | 3.14          | 4.14          | 4.43          | 4.00          | 4.00          |
| LastLayer          | 5.43          | 4.14          | 3.86          | 5.29          | 4.96          |
| AvgLayer           | 5.29          | 5.57          | 5.14          | 5.43          | 5.00          |

**Fig. 6:** Mean rank per benchmark (★ best). Concat 1st on 4/5, AMDF 1st on depth. Backbones: DINOv3-S/B/L · MAE-L · SigLIP2-L · V-JEPA2.1-L · PESpatial-L.

## Accuracy vs. parameters



**Fig. 7:** ViT-L, all 5 tasks: ≈19× (pose head larger: ≈22× on dense tasks).